Introduction to Computer Graphics 2025/2026

Intermediate Delivery

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A Little Walk

A low-poly 3D exploration game built with Three.js

GitHub: github.com/goncalooliveirasilva/a-little-walk
Live demo: a-little-walk.vercel.app

Goals

A Little Walk is a peaceful exploration game where the player controls a small animal wandering through an open world. The focus is on relaxation and discovery rather than challenge. The aim is to craft a beautiful environment with an expressive cartoon aesthetic, with ambient creatures (e.g., butterflies, bees, etc.), and varied elements including flowers, trees, vegetation, water areas, and ruins.

Current state of the project.

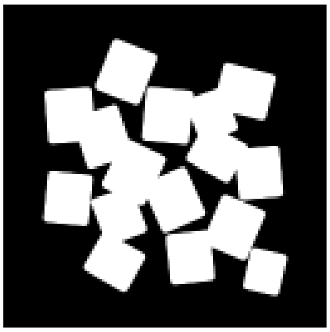
What is already done

- **Character:** a fox with walk, run, and idle animations, a third-person following camera, and player controls
- **World:** environment with a large ground plane, some trees, bushes, and rocks
- **Grass:** custom shader generating thousands of wind-animated grass blades using a noise texture, with a separate texture controlling where grass grows
- **Atmosphere:** gradient skydome shader (deep blue at the top, warm sunset tones at the horizon) and fog that hides the world's edges

What remains to be done

- **Terrain variation:** heightmap-based terrain introducing hills and valleys to replace the current flat ground plane
- **Water:** lakes in some parts of the map
- **Day/night cycle:** animated sun position and color, with the skydome, fog, and lighting responding dynamically throughout the cycle
- **Ambient wildlife:** butterflies, bees, and birds with simple movement patterns to bring life to the environment
- **More vegetation**: flowers and additional vegetation types to enrich the environment
- **Points of interest:** a few handcrafted points of interest, such as ancient ruins
- **Custom character model:** replace the placeholder fox asset with an original character modelled and rigged from scratch in Blender
- **Sound:** ambient sounds (wind, birds, footsteps, etc.)

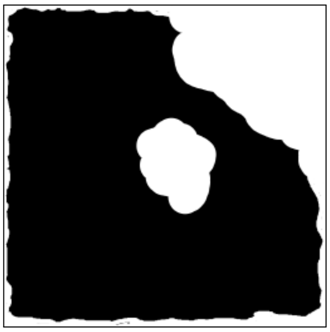
Some Technical Details



Bush/tree leaves texture

This texture is mapped onto flat planes that are randomly placed in a sphere. The fragment shader samples the alpha channel and discards pixels below a threshold, so the black background disappears and only the leaf shapes remain. (Which is better for performance than having thousands of leaves.)

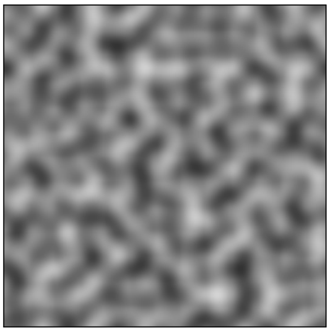
Made by myself in Canva.



Map grass texture

This texture is sampled at each blade's position. If the sampled value is lower than a specified threshold, the position of the vertex is set to (0, 0, 0). This way, in white areas we can see grass, and in black areas the grass is actually hidden in the origin point of the plane, so we cannot see it.

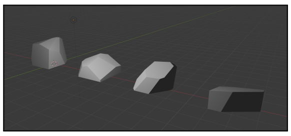
Made by myself in Krita.



Perlin noise (wind effect)

This texture is sampled in the vertex shader using `position.xz + uTime * windSpeed`, so here we are not only sampling with the position, but also with the time (clock time since the game started) multiplied by the windSpeed (a value we can customize and pass to the shader) and that produces the movement simulating the wind on the grass and bushes.

Made in this website.



3D Models

Trees and rocks were modelled in Blender and exported as GLTF. Each model is rendered through `InstancedMesh` for better performance. Trees and rocks were modelled by myself. The Fox model is not mine, as referenced in the project README.

Acknowledgments

To develop this initial version of the project, for the grass and bushes, I adapted techniques from this video[1]. Implementing custom shaders is something not explored in the classes, but I found them to be the best way to bring my ideas to life. I had only some basic prior contact with shaders, so I built some simpler parts myself and used AI for more complex ones, always making sure I understood the code and the logic rather than blindly trusting it to work. It has been a really fun learning experience. I also used AI to help build the camera movement and the loading page.

Some other concepts such as how to structure bigger projects, physics or how to deal with external animated models, I learned from this online course[2].

[1] `https://www.youtube.com/watch?v=cesPK0kYkyE&t=360s`
[2] `https://threejs-journey.com/`